Comparative Evaluation of Hybrid Models for Telecom Customer Churn Prediction: Discrimination, Calibration, and Error Analysis

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ABSTRACT

Customer churn prediction can help subscription businesses identify accounts that may discontinue a service, but a classifier selected only by accuracy can conceal missed churners and unnecessary retention contacts. This study compares individual classifiers and hybrid candidates using the public IBM Telco Customer Churn benchmark, which contains 7,043 customer records and 19 original predictor fields. The workflow includes data cleaning, exploratory analysis, twelve engineered features, leakage-aware preprocessing, six baseline classifiers, an equal-weight Random Forest–XGBoost probability ensemble, and five additional hybrid candidates. A stratified 80/20 split with random seed 42 was used, followed by five-fold stratified cross-validation on the training partition. Evaluation covers classification, ranking, agreement, probability quality, and threshold sensitivity. In training cross-validation, Random Forest obtained the highest mean ROC-AUC (0.8459), whereas Logistic Regression had the highest mean PR-AUC (0.6610). On the previously inspected holdout, weighted soft voting produced the highest PR-AUC among the five new hybrid candidates (0.6573), while stacking achieved their highest F1-score (0.6387). The paired bootstrap interval for the weighted-vote PR-AUC difference versus the existing RF–XGBoost hybrid included zero, and no McNemar comparison remained significant after correction. The results therefore support metric-specific comparison rather than a universal winning model. Author-supplied dashboard screenshots also document exploratory analysis, model evaluation, SHAP/permutation-importance views, and individual and batch prediction interfaces. Because the holdout was examined during development, the reported test-set findings are exploratory and require independent or temporal validation before deployment claims.

Keywords: customer churn; telecommunications; binary classification; hybrid ensemble; Random Forest; XGBoost; PR-AUC; probability calibration; SHAP; threshold analysis.

1. INTRODUCTION

1.1 Background

Telecommunications providers operate subscription services in which customer relationships generate recurring revenue. When a subscriber leaves, the provider may lose future revenue and incur additional acquisition costs to replace that account. Customer relationship management systems record information about tenure, service subscriptions, billing, contract type and payment behavior. These attributes can be analyzed to estimate which accounts resemble customers who previously churned. A churn-risk score can help an analyst prioritize review, but it is not a guarantee that an individual will leave. The value of a prediction depends on both its statistical quality and the way a business uses it.

The task is a binary classification problem: the target is whether a customer is recorded as having churned. The benchmark used here is a commonly circulated IBM Telco Customer Churn sample. It provides a compact, reproducible setting for comparing models and communicating their limitations. Its static records do not represent a live provider, and its results should not be read as a forecast of current market churn.

1.2 Problem Statement

A single headline metric does not describe all the errors that matter in churn prediction. A model may achieve high accuracy while failing to identify a substantial share of churners; another may detect more churners but generate more false alarms. Hybrid models can improve one metric by a small amount, yet that difference may be within normal validation variability. The practical problem addressed in this paper is how to compare baseline and hybrid classifiers using a shared evaluation protocol that reports ranking, threshold-dependent errors, probability quality and uncertainty together.

1.3 Research Gap

Recent churn studies examine ensembles, neural networks, feature selection and explainability, but their datasets and validation protocols differ. Consequently, performance values published by separate studies cannot be ranked as though they came from one common experiment. Reviews also identify class imbalance, interpretability, concept drift and business-aligned evaluation as recurring issues [11], [12], [16], [17]. This project does not claim that these topics have been ignored; instead, it brings several of them into one documented benchmark comparison and makes the uncertainty around small hybrid gains explicit.

1.4 Research Objectives

1. Build a leakage-aware preparation and feature-engineering pipeline.
2. Compare six baseline classifiers using a common split and metrics.
3. Compare RF–XGBoost voting with stacking and four other hybrid candidates.
4. Evaluate classification, ranking, calibration, thresholds and uncertainty.
5. Provide a dashboard for EDA, scoring, batch prediction and explanations.

1.5 Research Questions

- RQ1: How do six baselines compare with the RF–XGBoost hybrid?
- RQ2: Do five additional hybrid candidates improve performance consistently?
- RQ3: How do threshold and calibration choices change model interpretation?
- RQ4: How can dashboard outputs support individual and batch risk review?

1.6 Contributions

- A reproducible Telco churn pipeline with fold-wise preprocessing.
- A shared comparison of six baselines and six hybrid pipelines.
- Joint reporting of ranking, classification, calibration and uncertainty.
- A dashboard for EDA, explanations and single/batch scoring.

1.7 Organization of the Paper

Section 2 reviews churn prediction research and explainability; Section 3 describes the dataset, feature preparation and models; Section 4 explains the experimental protocol and metrics; Section 5 reports the saved results and dashboard evidence; Section 6 discusses their meaning; Section 7 records limitations; and Section 8 concludes with next steps.

2. RELATED WORK

2.1 Existing Approaches

Early churn systems commonly used statistical classifiers such as logistic regression and decision trees because their outputs can be inspected and their training cost is manageable. Logistic regression gives a direct linear reference, while a single tree can represent non-linear feature rules but may be unstable when the data change. Random Forest extends a tree through bagging and randomized feature selection, while gradient boosting builds trees sequentially to reduce residual error. Beeharry and Fokone [1] studied a hybrid voting approach for telecom churn, and Edwine et al. [3] compared optimized Random Forest, SVM and KNN models with feature selection and controlled undersampling. These studies show that model choice cannot be separated from data preparation: class balance, feature selection and validation design influence the apparent ranking. The current experiment therefore retains a linear model, tree learners and a kernel method as distinct reference points before assessing hybrid combinations.

2.2 Machine Learning / Computational Approaches

Recent work continues to compare tree ensembles, boosting methods and hybrid classifiers. Prabadevi et al. [6] evaluated stochastic gradient boosting, Random Forest, KNN and logistic regression, while Wagh et al. [8] examined several classifiers for telecom data. Zhou et al. [5] framed the task as an early-warning problem for enterprise accounts, emphasizing the timing of a retention response rather than only retrospective classification. Poudel et al. [10] paired tabular classifiers with global and local explanations. Manzoor et al. [11] reviewed the churn workflow and identified business-facing evaluation as an important concern. Imani et al. [16] synthesized 61 studies in detail from a broader PRISMA-guided set of 240 papers and discussed class imbalance, interpretability and concept drift. Across this work, Random Forest and boosting are common choices for structured data, but reported leaders vary with the customer population, class ratio and threshold. For that reason, the current study reports both ROC-AUC and PR-AUC, alongside threshold-based measures, rather than treating a single accuracy value as a complete comparison.

2.3 Advanced / Alternative Approaches

Alternative approaches include convolutional networks and distributed processing [2], topological data analysis [14], segmented hybrid models [9], and feature-optimization ensembles [18]. Tariq et al. [2] combine a CNN with Apache Spark, a design aimed at distributed processing rather than a lightweight CPU comparison. Sagming et al. [14] introduce topological data analysis as another way to represent customer structure. De Caigny et al. [9] examine segmented black-box models and interpretability across multiple churn datasets, while Syah and Elveny [18] combine feature optimization with weighted boosting in e-commerce. Suh [7] studied operational data from a home-appliance rental business and reported strong F-measure and AUC results in that setting. Recent telecom work has also combined boosted learners, feature selection and resampling [15], or used stacked learners with SHAP analysis [19], [20]. These designs broaden the available methodological options, but some require more data, more compute or additional data structures than the tabular workflow used here. Their metrics are still useful context, not direct benchmarks, because the sample sizes, class ratios, feature definitions and validation schemes are not identical.

2.4 Explainability / Interpretability (or other relevant sub-theme)

Interpretability has become a more visible part of churn research. SHAP-based explanations have been applied to tabular telecom classifiers [9], [10], [15], [19], [20], while some studies pair model output with dashboards or visualization applications [13]. SHAP can summarize how input features contribute to a prediction at global and local levels; LIME provides a local surrogate view. These tools can make a model easier to inspect, but the explanation remains conditional on the fitted model and available input data. It does not establish that a feature causes a customer to churn. Probability quality is a related but separate issue: a model may rank customers well while its scores are too high or too low to be interpreted as calibrated risk [24]. Generalizability is also constrained when studies rely on one static data source. The present work includes author-supplied SHAP and permutation-importance screenshots, but interprets feature rankings as model associations rather than causal findings and notes the need for external validation.

2.5 Critical Comparison

Table 1 groups twenty studies selected from the project literature survey. The studies are compared by their methodological contribution and relevance to the present problem rather than by ranking their published accuracy or AUC values. Those values were measured on different datasets and under different validation protocols, so a direct numeric leaderboard would be misleading. The limitation column describes scope or transfer boundaries relevant to this project; it is an analytical reading of each study and is not necessarily a limitation stated verbatim by the original authors.

#	Study / Publication	Key Contribution	Limitation / Relevance
1	Beeharry & Fokone, Concurrency Comput. Pract. Exp., 2022 [1]	Hybrid voting approach for telecom churn; evaluates combinations of machine-learning learners across customer datasets.	The result is specific to its datasets and experimental setup; it does not settle calibration or threshold choice for this project.
2	Tariq et al., J. Model. Manag., 2022 [2]	Uses a two-dimensional CNN and Apache Spark for distributed churn processing on a Telco Customer Churn dataset.	The CNN/distributed design has greater implementation cost than a compact tabular baseline comparison.
3	Edwine et al., Math. Probl. Eng., 2022 [3]	Combines mutual-information feature selection, controlled-ratio undersampling and tuned RF, SVM and KNN models.	The reported results depend on two telecom datasets and a specific undersampling/optimization protocol.
4	Peng et al., PLOS ONE, 2023 [4]	Studies customer churn prediction and model interpretability in the banking context.	The banking population and feature definitions differ from the telecom benchmark used here.
5	Zhou et al., PLOS ONE, 2023 [5]	Develops an ensemble-learning early-warning method for enterprise telecom customer churn.	The enterprise-customer setting may not transfer directly to individual residential subscriptions.
6	Prabadevi et al., Int. J. Intell. Netw., 2023 [6]	Compares stochastic gradient boosting, Random Forest, KNN and logistic regression for customer churn analysis.	The study emphasizes classifier comparison; the present paper additionally reports probability calibration and paired uncertainty.
7	Suh, J. Big Data, 2023 [7]	Uses operational water-purifier rental data and evaluates churn models with F-measure and AUC.	The rental-service domain and proprietary operating data differ from the public telecom sample.
8	Wagh et al., Results Control Optim., 2024 [8]	Examines machine-learning classifiers for telecom churn and discusses churn-related customer behavior.	Published scores depend on the source data and resampling protocol and are not directly comparable to this experiment.
9	De Caigny et al., Decis. Support Syst., 2024 [9]	Extends hybrid segmented modelling with black-box learners and SHAP-based interpretation; benchmarks fourteen churn datasets.	The segmented hybrid framework is more specialized than the single-population pipelines used in this project.
10	Poudel et al., Mach. Learn. Appl., 2024 [10]	Compares tabular telecom classifiers and combines predictive performance with global and local explanations.	The findings reflect its dataset and validation design; probability calibration is a separate concern.
11	Manzoor et al., IEEE Access, 2024 [11]	Reviews machine-learning churn methods and provides guidance for business practitioners across workflow stages.	As a review, it synthesizes reported work rather than benchmarking one implementation on the current dataset.
12	Barsotti et al., SN Comput. Sci., 2024 [12]	Surveys a decade of telecom churn methods, datasets, metrics and open research challenges.	The survey highlights heterogeneous protocols, which limit direct ranking of published performance values.
13	Singh et al., Data Sci. Manag., 2024 [13]	Combines banking churn model comparison with an R Shiny visualization application.	The banking domain differs from telecom and its interface is not a direct evaluation of this project dashboard.
14	Sagming et al., J. Big Data, 2024 [14]	Explores topological data analysis together with machine learning for customer churn prediction.	Topological representation adds complexity and is not part of the current tabular feature pipeline.
15	Asif et al., Results Eng., 2025 [15]	Proposes XAI-Churn TriBoost using XGBoost, CatBoost and LightGBM soft voting with SHAP and LIME on a large telecom dataset.	The study uses a dataset of a different scale and experimental setup; its headline scores should not be compared directly.
16	Imani et al., Mach. Learn. Knowl. Extr., 2025 [16]	PRISMA-guided review maps 240 churn papers and synthesizes 61 studies in detail, identifying imbalance, drift and interpretability issues.	The authors report study heterogeneity and do not conduct a quantitative meta-analysis.
17	Shahabikargara et al., J. Inf. Telecommun., 2026 [17]	Surveys churn analysis across customer journeys, features and sectors including telecom, banking and gaming.	Its broad cross-industry coverage is contextual rather than a controlled benchmark on the same dataset.
18	Syah & Elveny, Decis. Anal. J., 2025 [18]	Combines PCA, feature optimization and weighted XGBoost/LightGBM voting with SHAP in an e-commerce setting.	The e-commerce dataset and optimization pipeline differ from the telecom benchmark and current computational scope.
19	Rasool et al., Math. Model. Eng. Probl., 2025 [19]	Uses a stacked ensemble of tree and boosting classifiers with SHAP analysis for telecom churn prediction.	The method is tied to its dataset and stacking design; external validation remains important.
20	El Attar & El-Hajj, Front. Artif. Intell., 2026 [20]	Compares seven models on the 7,043-record Telco dataset and combines soft voting, SHAP, segmentation and threshold analysis.	The evaluation protocol and model configuration differ from this implementation; published values are not a direct replication.

Table 1. Selected customer-churn studies, their methodological contributions and project-relevant scope limits.

2.6 Research Gap (Summary)

The literature demonstrates that linear, tree-based, boosting, neural and hybrid models have all been applied to churn prediction. It also shows that the reported performance depends heavily on the source population, target definition, class balance and validation procedure. This project addresses the comparison problem by evaluating six individual classifiers and six hybrid pipelines under one local protocol, then examining threshold sensitivity, calibration and paired uncertainty. The resulting contribution is a transparent benchmark and working dashboard workflow, not a new learning algorithm.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

The implementation is organized into three connected regions, as shown in Figure 1. The offline experimentation region loads the Telco CSV, checks its schema and target, cleans the charge field, performs exploratory analysis, creates a stratified 80/20 split and fits a leakage-safe preprocessing pipeline. The training partition supports five-fold stratified cross-validation and fitting of six baseline classifiers, the Random Forest–XGBoost hybrid and five additional hybrid candidates. Holdout evaluation records classification metrics, calibration, threshold behaviour, subgroup errors and paired comparisons. The saved-artifact region stores models, figures, reports, predictions and metrics for reuse. The interactive Streamlit region uses the persisted scoring model and exposes six pages, including single-customer and batch CSV prediction. The five extra hybrid candidates and the separate calibration and threshold analyses remain offline evaluation outputs rather than separate dashboard models.

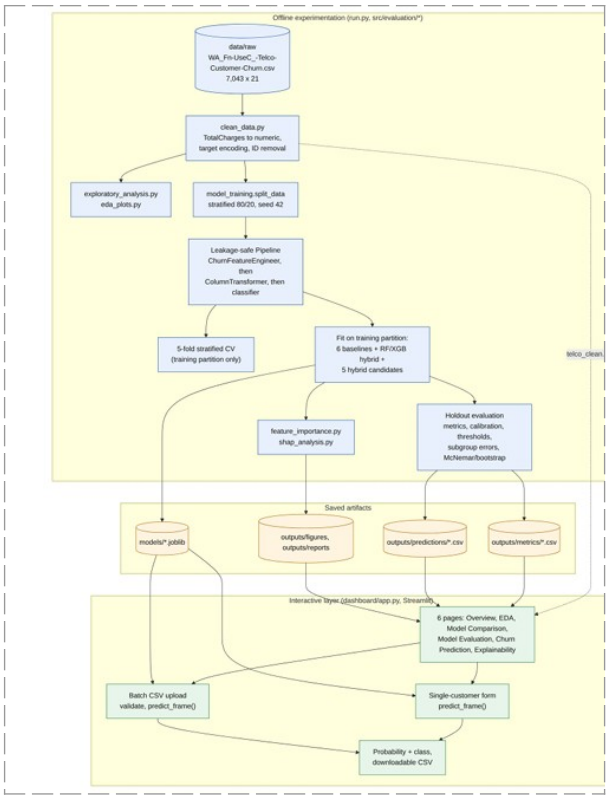


Fig. 1. Repository-traced architecture of the implemented churn system. Blue denotes offline experimentation, orange denotes saved artifacts, and green denotes the interactive Streamlit layer (author-supplied diagram, rendered from fig1_system_architecture.mmd).

3.2 Dataset(s)

One public tabular benchmark was used. It is suitable for a reproducible academic comparison because its fields cover contract, tenure, service adoption, payment method and billing, while the target is already defined as a binary churn label.

Dataset profile: 7,043 customers; 5,174 non-churners (73.46%) and 1,869 churners (26.54%); 21 raw columns; 19 predictors after excluding customerID and Churn; 11 blank TotalCharges entries before cleaning; zero exact duplicate rows. The local CSV SHA-256 recorded in the project draft is 16320c9c1ec72448db59aa0a26a0b95401046bef5d02fd3aeb906448e3055e91.

- IBM Telco Customer Churn sample: 7,043 rows and 21 raw columns, including 19 predictor fields, customerID and the Churn target. The target contains 1,869 churn records (26.54%) and 5,174 non-churn records (73.46%).
- No second dataset was used. The three-way dataset comparison discussed in some literature is not part of this implementation; all reported experiments in this paper use the same Telco CSV.

3.3 Data Preprocessing

Preprocessing combines fixed cleaning rules with transformations learned inside the model pipeline. Fixed cleaning is applied before the train/test split; the imputer, scaler and encoder are fitted on training rows only and are re-fitted within each cross-validation fold.

- The eleven blank TotalCharges strings were converted to numeric values and set to 0.0 because all eleven rows had tenure equal to zero. Exact duplicate rows were not found. CustomerID was removed from the predictor matrix, and Churn was encoded as 1 for Yes and 0 for No.
- The IQR report recorded no outliers in tenure, MonthlyCharges or TotalCharges under the project rule, so no observations were removed on that basis. The cleaning report found no duplicate customer IDs.
- Numeric columns use median imputation and standard scaling; categorical columns use most-frequent imputation and one-hot encoding. These transformations are wrapped in scikit-learn pipelines to reduce leakage.

3.4 Exploratory Analysis and Feature Engineering

EDA compares churn proportions across contract type, payment method, internet service, tenure and senior-citizen status, and uses distributions and box plots to inspect charges. The supplied contract view shows churn rates of about 42.7% for month-to-month customers, 11.3% for one-year contracts and 2.8% for two-year contracts. The supplied EDA views also show higher observed churn for fiber-optic users and electronic-check payers. These are descriptive associations, not causal effects. Twelve features were engineered: tenure group, service counts, protection-service count, streaming-service count, average monthly charge, charge delta, charges per service, month-to-month flag, automatic-payment flag, electronic-check flag, family flag and a month-to-month by fiber-optic interaction.

The twelve derived fields are tenure_group, service_count, protection_service_count, streaming_service_count, avg_monthly_charge, charge_delta, charges_per_service, is_month_to_month, auto_payment, is_electronic_check, has_family and mtm_fiber. After categorical encoding, the model pipeline receives 61 columns. Some engineered indicators overlap with one-hot encoded raw categories, so feature attribution can be distributed across correlated inputs.

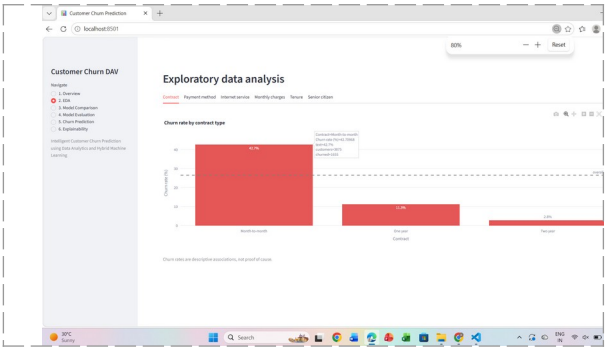


Fig. 2. Author-supplied exploratory-analysis view showing churn rate by contract type (screen capture, 25 October 2026).

3.5 Proposed Model(s) and Baselines

The experiment uses six baseline classifiers and several ensemble candidates. Baselines provide contrasting assumptions and model capacity; all are evaluated under the same train/test partition and metric definitions.

- Logistic Regression provides a linear reference model. A Decision Tree offers a compact nonlinear rule-based baseline, while Random Forest averages decorrelated trees to reduce variance.
- XGBoost builds boosted decision trees sequentially and is included as a strong tabular learner [22]. Its probability output is also paired with Random Forest in the project's two-model hybrid.
- RBF-SVM tests a kernel-based margin classifier, while the MLP tests a feed-forward neural model. The hybrid set adds equal-weight RF–XGBoost soft voting, a weighted RF/XGBoost/SVM vote, stacking, chi-square feature selection with an ensemble, RF feature selection with ANN, and XGBoost leaf embedding with ANN.

3.6 Algorithmic / Mathematical Formulation

The existing RF–XGBoost hybrid averages the class-probability estimates of its two members. The candidate weighted vote assigns larger weight to XGBoost. A class label is produced only after a threshold is chosen.

$$p_{hybrid}(x) = 0.5\ p_{RF}(x) + 0.5\ p_{XGB}(x) \quad (1)$$

$$p_{weighted}(x) = 0.2\ p_{RF}(x) + 0.6\ p_{XGB}(x) + 0.2\ p_{SVM}(x) \quad (2)$$

$$\hat{y}(x) = 1 \text{ if } p(x) \geq t; \text{ otherwise } \hat{y}(x) = 0 \quad (3)$$

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

The project recommends Python 3.12 and uses pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0, XGBoost 3.1.3, SHAP 0.50.0, Matplotlib 3.10.8, Seaborn 0.13.2, Plotly 6.5.2, Streamlit 1.64.0 and joblib 1.5.3 in the supplied requirements file. The workflow is CPU-oriented and does not require a GPU. The exact laptop/CPU/RAM used for the screenshot capture was not recorded, so no hardware benchmark is claimed.

4.2 Dataset Split / Design

The data was divided using a stratified 80/20 split with random_state=42: 5,634 customers for training and 1,409 for the holdout. The holdout contains 374 churners. Five-fold StratifiedKFold with shuffling and seed 42 was run on the training partition. Imputation, scaling and one-hot encoding were learned inside the pipeline for each fold.

4.3 Hyperparameter Configuration

Baseline settings were fixed for a controlled academic comparison rather than tuned exhaustively. The project documentation records a depth-5 decision tree with minimum leaf size 20, a 300-tree Random Forest with maximum depth 10 and minimum leaf size 5, and XGBoost with 300 estimators, learning rate 0.05, depth 4, subsample 0.8 and column sampling 0.8. The RF–XGBoost hybrid uses equal weights by default. The weighted soft-voting candidate uses RF/XGBoost/SVM weights of 0.2/0.6/0.2. A search routine exists in the repository, but no completed tuning run is reported in the archived results.

4.4 Validation Strategy

Five-fold stratified cross-validation provides a training-only estimate of ranking and classification stability. Paired bootstrap resampling of the holdout was used for selected PR-AUC differences, and paired McNemar tests were calculated for classification errors. Since the holdout was consulted during candidate development and diagnostics, these checks do not turn it into a fresh confirmatory test set.

4.5 Evaluation Metrics

- Accuracy, precision, recall/sensitivity, specificity and F1 describe threshold-dependent classification. Precision measures the share of flagged customers who churn; recall measures the share of true churners detected; specificity measures correctly identified non-churners. Higher is better for these metrics, but trade-offs matter.
- ROC-AUC measures ranking across thresholds, while PR-AUC (average precision) emphasizes the precision–recall relationship for the positive class. MCC and Cohen's kappa summarize agreement while accounting for both classes. Higher values indicate stronger ranking or agreement.
- Brier score, log loss and expected calibration error (ECE) assess probability quality; lower values are preferred. A model may have strong ROC-AUC but still produce probabilities that are too high or poorly calibrated. All metrics in this paper treat churn as the positive class.

4.6 Statistical Testing

Paired bootstrap intervals were computed for selected PR-AUC differences. The weighted soft-voting candidate exceeded the RF–XGBoost hybrid by 0.0063 PR-AUC, but the 95% interval ranged from −0.0005 to 0.0130 and included zero. The stacking difference was −0.0046 with a 95% interval from −0.0110 to 0.0014. Four McNemar comparisons were assessed with a Bonferroni threshold of 0.0125; none remained significant. These results do not establish a statistically reliable winner.

5. RESULTS

The following values are saved empirical outputs from the supplied Stage 8 project archive. They are exploratory holdout results, not independent confirmation, because the holdout was used in model-development and diagnostic comparisons. Tables use the archived CSV metrics as the numerical source of record.

5.1 Overall Performance

On the holdout, the MLP obtained the highest accuracy (0.7977) but had the lowest recall among the six individual classifiers (0.5588). Random Forest had the highest baseline F1 (0.6344); the equal-weight RF–XGBoost hybrid recorded ROC-AUC 0.8439 and PR-AUC 0.6510. In five-fold training cross-validation, Random Forest had the highest mean ROC-AUC (0.8459 ± 0.0102), while Logistic Regression had the highest mean PR-AUC (0.6610 ± 0.0139). These differences show that rankings depend on the metric.

Cross-validation summary: Random Forest achieved mean ROC-AUC 0.8459 (SD 0.0102); the RF–XGBoost hybrid reached 0.8450 (SD 0.0116). Logistic Regression achieved mean PR-AUC 0.6610 (SD 0.0139), compared with 0.6590 (SD 0.0240) for Random Forest and 0.6582 (SD 0.0236) for the RF–XGBoost hybrid. The differences are small relative to fold variability.

5.2 Model / Method Comparison

Table 2 reports the seven baseline pipelines, including the existing RF–XGBoost hybrid. Accuracy, F1 and ranking metrics are shown together because no one measure captures all churn-related errors.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.7353	0.5008	0.7888	0.6127	0.8420	0.6370
Decision Tree	0.7488	0.5181	0.7674	0.6185	0.8321	0.6244
Random Forest	0.7693	0.5476	0.7540	0.6344	0.8413	0.6462
XGBoost	0.7530	0.5233	0.7807	0.6266	0.8428	0.6498
RBF-SVM	0.7828	0.5833	0.6364	0.6087	0.8249	0.6142
MLP	0.7977	0.6353	0.5588	0.5946	0.8365	0.6230
RF + XGBoost hybrid	0.7644	0.5396	0.7647	0.6327	0.8439	0.6510

Table 2. Baseline classifiers and the existing RF–XGBoost hybrid on the exploratory holdout; values are from archived project CSV outputs.

Hybrid candidate	CV ROC-AUC	CV PR-AUC	Holdout F1	Holdout ROC-AUC	Holdout PR-AUC
Stacking (RF + XGB + SVM → LR)	0.8455	0.6586	0.6387	0.8419	0.6464
Weighted soft voting (RF/XGB/SVM)	0.8440	0.6577	0.6253	0.8443	0.6573
Chi-square + ensemble	0.8402	0.6448	0.6216	0.8342	0.6464
RF feature selection + ANN	0.8399	0.6436	0.5714	0.8406	0.6379
XGB leaf embedding + ANN	0.8311	0.6384	0.5662	0.8291	0.6206

Table 3. Results for the five additional hybrid candidates; holdout values are exploratory.

Model	PR-AUC	Brier (↓)	Log loss (↓)	ECE (↓)
Weighted soft voting	0.6573	0.1496	0.4508	0.0964
Existing RF+XGBoost hybrid	0.6510	0.1555	0.4665	0.1171
XGBoost	0.6498	0.1596	0.4777	0.1232
Stacking	0.6464	0.1663	0.5041	0.1566
Random Forest	0.6462	0.1537	0.4623	0.1110
Logistic Regression	0.6370	0.1672	0.4950	0.1495
ANN	0.6230	0.1401	0.4267	0.0244
SVM	0.6142	0.1446	0.4470	0.0487

Table 4. Probability-quality measures for selected models on the exploratory holdout.

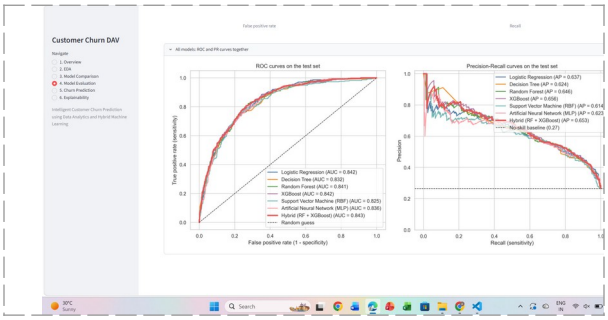


Fig. 3. Author-supplied dashboard capture showing ROC and precision–recall curves for the evaluated classifiers (screen capture, 25 October 2026).

Figure note: dashboard screenshots were supplied by the authors and were not generated for this paper. The visible dashboard values in these screenshots differ slightly from some archived Stage 8 CSV values, consistent with separate execution snapshots. For reproducible numerical tables, the archived CSV files in the project bundle are used as the source of record.

5.3 Error Analysis and Threshold Sensitivity

For the archived RF–XGBoost holdout predictions, the confusion matrix contains TN = 791, FP = 244, FN = 88 and TP = 286. At threshold 0.50, recall is 0.7647 and specificity is 0.7643. Lowering the threshold to 0.30 increases recall to 0.8877 and reduces false negatives from 88 to 42, but false positives rise from 244 to 406 and specificity falls to 0.6077. The threshold therefore changes the balance between missed churners and unnecessary retention contacts; a business cost model is needed before selecting an operating point.

5.4 Explainability and Feature Importance

The supplied explainability screenshot shows permutation importance and a SHAP summary for the RF–XGBoost model. Contract type and tenure are among the highest-ranked original fields in permutation importance. In the encoded SHAP summary, the month-to-month indicator, tenure, the month-to-month/fiber interaction, contract indicators, online-security status and charge variables appear among the strongest contributors. These attributions explain the fitted model's output and should not be interpreted as causal drivers of churn.

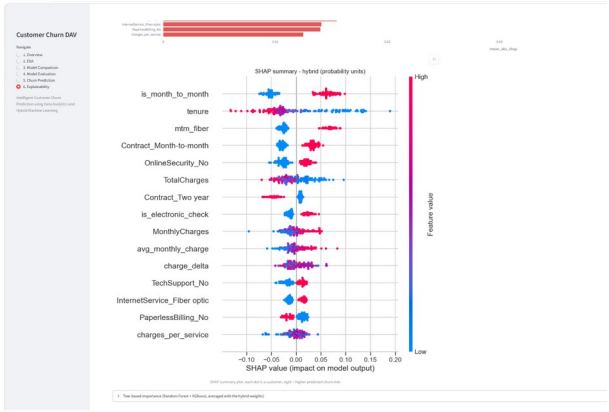


Fig. 4. Author-supplied explainability page showing permutation importance, mean absolute SHAP values and a SHAP summary plot (screen capture, 25 October 2026).

5.5 Dashboard Prediction Outputs

The supplied Streamlit screenshots show six navigation pages: Overview, EDA, Model Comparison, Model Evaluation, Churn Prediction and Explainability. In the single-customer example, the entered profile produced a churn probability of 78.6% and the interface displayed a CHURN label using a 0.50 cutoff. In the batch example, a five-row CSV was scored and three records were flagged. These figures document example interface outputs; they do not measure dashboard usability or retention impact.

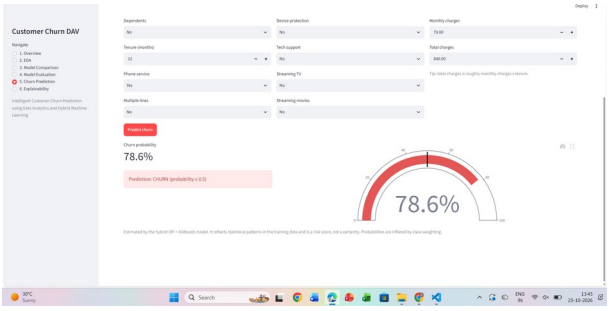


Fig. 5. Single-customer prediction page showing an example 78.6% churn probability and threshold-based label (author-supplied screen capture, 25 October 2026).

The batch workflow accepts a customer-feature CSV, validates required columns, produces churn_probability and churn_prediction fields, and offers the scored output for download. The image below records one sample batch run.

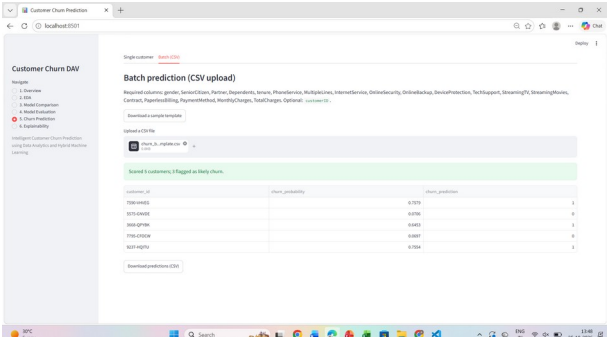


Fig. 6. Batch CSV prediction output for a five-customer example, with three records flagged as likely churn (author-supplied screen capture, 25 October 2026).

6. DISCUSSION

6.1 Interpretation of Findings

The MLP's high accuracy is accompanied by lower recall, which is consistent with a classifier that is more conservative about predicting the positive class at the fixed 0.50 threshold. Random Forest and XGBoost capture nonlinear interactions in the tabular attributes, while the RF–XGBoost average combines their probability estimates. The gain from combining them is small because the base learners make correlated predictions. Class weighting raises sensitivity but also increases false-positive counts and can distort probability calibration.

6.2 Comparison With Previous Studies

The results are consistent with studies that find tree ensembles and boosting useful for structured churn data [1], [6], [8], [10], [15], [20]. They also support the survey findings that no single algorithm is universally best across domains and datasets [11], [12], [16], [17]. Unlike work that reports one best accuracy value, the current experiment shows different leaders for accuracy, F1, cross-validation ROC-AUC, cross-validation PR-AUC and calibration. Since published studies use different datasets and protocols, this paper compares methodological direction rather than raw scores across papers.

6.3 Theoretical Implications

Churn prediction should be treated as a decision-support problem rather than only a classification leaderboard. Researchers should report the positive-class definition, prevalence, split strategy, threshold and error counts, and should assess probability quality when scores are intended to represent risk. Small ensemble improvements require uncertainty estimates and independent validation before being described as robust.

6.4 Practical Implications

A retention analyst can use the dashboard to inspect customer distributions, view model comparisons, score a single record and upload a batch CSV. The single-customer screenshot shows an illustrative 78.6% churn probability for one form input; the batch example scores five records and flags three. These are examples of interface operation, not evidence that a retention offer will prevent churn or generate a financial return. In practice, thresholds should be selected using contact capacity, offer cost and the relative cost of false negatives and false positives.

6.5 Unexpected Findings

The model with the highest accuracy did not have the highest recall or F1. The weighted soft-voting candidate had the best PR-AUC among the five additional hybrids, while stacking had the best F1 among those candidates. The paired uncertainty intervals included zero, so the apparent ordering is not strong evidence of a stable advantage.

7. LIMITATIONS

- The 20% holdout was inspected during candidate development and diagnostics. Its reported values are exploratory; an untouched external or temporal test set is needed for confirmation.
- Only one static public sample was used. The data contain no time series, intervention records, customer lifetime value or retention-offer outcomes.
- Class weighting affects recall and probability calibration. Calibration metrics are reported, but a final calibration model and cost-based operating threshold were not selected.
- The literature and the benchmark use heterogeneous providers and datasets. The model's feature attributions are associative, and subgroup differences have not been validated with confidence intervals or an external sample. The supplied screenshot documents the SHAP view, but the archive does not contain a separate saved SHAP attribution export for independent reproduction.

8. CONCLUSION AND FUTURE WORK

This paper compared six individual classifiers, an RF–XGBoost probability ensemble and five additional hybrid candidates for telecom customer churn prediction. The study combined exploratory analysis, feature engineering, stratified cross-validation, holdout metrics, threshold sensitivity, probability-quality measures and dashboard evidence. The saved results show metric-specific trade-offs rather than a single consistently superior classifier. The contribution is an auditable benchmark workflow and a dashboard that presents churn analysis and sample predictions.

Future work should first reserve a new independent or time-ordered dataset for confirmation, then complete training-only hyperparameter search and select thresholds using a documented retention cost matrix. Probability calibration and subgroup uncertainty analysis should be evaluated on validation data. The dashboard and SHAP workflow should be rerun in a clean environment with saved artifacts and version metadata.

DECLARATIONS

Funding

No external funding was received for this student project.

Conflict of Interest

The authors declare no financial or personal conflict of interest related to this academic project.

Ethics Statement

The study uses a public secondary benchmark dataset and involves no new data collection from human participants. No separate institutional ethics approval was documented. Customer identifiers were excluded from model predictors and are not reported as research findings.

Data Availability

The CSV used in the project is included in the supplied project archive. The public IBM Telco Customer Churn sample source is listed in Reference [25]. The local file checksum is included in Section 3.2 for reproducibility.

Author Contributions

Allipuram Akhil and Devineni Surya jointly contributed to project design, data preparation, implementation, evaluation, interpretation and manuscript development. The authors should confirm the final allocation of individual tasks before submission.

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